Overshoot Elimination in a Series RLC Circuit via Analog PID Control

Electronic Engineering Technical Documentation

Project Logbook

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Abstract

This document describes the design, classical control analysis, and SPICE simulation of an analog PID controller applied to a series RLC circuit, with the goal of eliminating the step-response overshoot inherent to its underdamped open-loop dynamics. This is the first of two related projects; the second applies the same PID design methodology to a buck converter (repository `BUCK_1` covers an earlier, IC-based adjustable buck design, not the PID-controlled version). Documented as an honest engineering logbook: every result is marked as **calculated**, **simulated**, or **pending** — never filled in with invented placeholder data. The plant is fixed and its open-loop overshoot confirmed both analytically (52.7%) and in SPICE (52.68%). The PID gains are derived via pole placement ($K_p \approx 27.87$, $K_i \approx 455{,}905$, $K_d \approx 3.114 \times 10^{-4}$) and the op-amp compensator topology is defined. **Still open**: the specific R/C component values realizing those gains, the full closed-loop schematic, and its SPICE validation — explicitly marked **PENDING**.

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1 Project Objective

Design and validate an analog PID controller, implemented with operational amplifiers, that closes the loop around a series RLC circuit and eliminates the overshoot present in its open-loop step response, using classical control design methods (transfer function derivation, root locus / pole placement). This project establishes the PID design methodology later applied to a discrete, analog-PID-controlled buck converter (a separate, subsequent project).

2 Plant: Series RLC Circuit

2.1 Design criteria

Component values were not assigned; they were chosen to satisfy two criteria: (1) clearly underdamped (ζ well below 1) so the open-loop overshoot is unambiguous, and (2) a natural frequency practical for breadboard construction and standard lab equipment (function generator + oscilloscope), avoiding both breadboard-parasitic-dominated frequencies (too high) and impractically long observation times (too low).

Table 1: Design targets and resulting component values

Parameter	**Value**	**Source**
Target natural frequency, f_0	5 kHz	Defined by design
Target damping ratio, ζ	0.2 (underdamped, ≈53 % overshoot target)	Defined by design
Capacitance, C	100 nF	Defined by design (standard value)
Inductance, L	10.13 mH	**Calculated**, Sec. 2.2
Resistance, R	127.32 Ω	**Calculated**, Sec. 2.2

2.2 Component value derivation

With $\omega_n = 2\pi f_0 = 2\pi(5000) \approx 31{,}416\,\text{rad/s}$, and the standard series-RLC relations:

$$L = \frac{1}{\omega_n^2 C} \tag{1}$$

$$R = 2\zeta\sqrt{L/C} \tag{2}$$

$$L = \frac{1}{(31416)^2(100\times10^{-9})} = 10.13\,\text{mH} \tag{3}$$

$$R = 2(0.2)\sqrt{\frac{10.13\times10^{-3}}{100\times10^{-9}}} = 127.32\,\Omega \tag{4}$$

3 Open-Loop Transfer Function

With the capacitor voltage V_C as the output and the source voltage V_{in} as the input, the standard second-order series-RLC transfer function is:

$$\frac{V_C(s)}{V_{in}(s)} = \frac{1/LC}{s^2 + \dfrac{R}{L}s + \dfrac{1}{LC}} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{5}$$

with $\omega_n = 1/\sqrt{LC}$ and $\zeta = \frac{R}{2}\sqrt{C/L}$, confirming the values imposed by design in Section 2.

3.1 Predicted open-loop overshoot

For an underdamped second-order system, the standard closed-form percent overshoot for a step input is:

$$\%OS = \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right) \times 100 \tag{6}$$

With $\zeta = 0.2$: $\%OS = \exp\left(\dfrac{-\pi(0.2)}{\sqrt{1-0.04}}\right) \times 100 \approx \boxed{52.7\%}$.

This confirms the plant has substantial, clearly-visible open-loop overshoot – a real problem for the PID controller (Section 4) to solve, not a marginal one.

3.2 SPICE confirmation

Simulated. The circuit was built directly in LTspice (series *R-L-C*, driven by a 1 V step via a `PULSE` source, V_C probed at the *L-C* junction, `.tran 2m`). The peak and final values were extracted numerically from the `.raw` binary file (not read visually from the plot):

Peak V_C	1.526,79 V, at $t = 102.28\,\mu s$
Final (settled) V_C	1.000,00 V
Overshoot	$(1.52679 - 1)/1 \times 100 = \boxed{52.68\%}$

This matches the analytical prediction (52.7%) to within 0.02 percentage points, validating both the transfer-function derivation and its SPICE implementation before any control is added.

4 PID Controller Design

4.1 Parallel vs. series PID form

Two classical structures exist for a PID compensator:

$$\text{Parallel (ideal):} \quad C(s) = K_p + \frac{K_i}{s} + K_d s \tag{7}$$

$$\text{Series (interacting):} \quad C(s) = K_c\left(1 + \frac{1}{T_i s}\right)(1 + T_d s) \tag{8}$$

The series form is what results naturally from cascading a PI stage into a PD stage (fewer op-amps), but its effective gains interact: expanding it gives $K_p = K_c(1 + T_d/T_i)$, $K_i = K_c/T_i$, $K_d = K_c T_d$ – changing T_i also moves the effective K_p. **The parallel form was chosen** for this design: each gain (K_p, K_i, K_d) is independently tunable, which maps directly onto the pole-placement method used below (Section 4.3) and is more straightforward to justify/defend gain-by-gain.

4.2 Why the closed-loop system is third order

The plant $G(s)$ (Section 3) is second order (two energy-storage elements, L and C). The controller contributes exactly one additional pole, at $s = 0$, from the integral term alone – the P and D terms add no poles. The full derivation, step by step:

Step 1 – combine $C(s)$ into a single fraction (common denominator s):

$$C(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_p s}{s} + \frac{K_i}{s} + \frac{K_d s^2}{s} = \frac{K_d s^2 + K_p s + K_i}{s} \tag{9}$$

Step 2 – multiply $C(s)G(s)$:

$$C(s)G(s) = \frac{K_d s^2 + K_p s + K_i}{s} \cdot \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2 \left(K_d s^2 + K_p s + K_i\right)}{s\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right)} \tag{10}$$

Step 3 – the characteristic equation is $1 + C(s)G(s) = 0$. Clearing the denominator (multiplying through by it):

$$s\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right) + \omega_n^2\left(K_d s^2 + K_p s + K_i\right) = 0 \tag{11}$$

Step 4 – expand each product separately. First term:

$$s\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right) = s^3 + 2\zeta\omega_n s^2 + \omega_n^2 s \tag{12}$$

Second term:

$$\omega_n^2\left(K_d s^2 + K_p s + K_i\right) = \omega_n^2 K_d s^2 + \omega_n^2 K_p s + \omega_n^2 K_i \tag{13}$$

Step 5 – add both expansions and group by power of s:

$$s^3 \text{ term:} \quad s^3 \tag{14}$$

$$s^2 \text{ terms:} \quad 2\zeta\omega_n s^2 + \omega_n^2 K_d s^2 = \left(2\zeta\omega_n + \omega_n^2 K_d\right) s^2 \tag{15}$$

$$s^1 \text{ terms:} \quad \omega_n^2 s + \omega_n^2 K_p s = \omega_n^2(1 + K_p)s \tag{16}$$

$$s^0 \text{ term:} \quad \omega_n^2 K_i \tag{17}$$

$$s^3 + \left(2\zeta\omega_n + \omega_n^2 K_d\right) s^2 + \omega_n^2(1 + K_p)s + \omega_n^2 K_i = 0 \tag{18}$$

The degree-3 result is not a choice – it is forced by combining a 2nd-order plant with a PID (which has exactly 3 free gains, matching the 3 non-leading coefficients of a cubic). The lone s left over from the integrator in Step 1 is what multiplies against the plant's s^2 in Step 4, producing s^3; without an integral term, $C(s)G(s)$ would only reach s^2 in the denominator and the closed loop would be 2nd order. This is precisely why full pole placement – choosing all three closed-loop poles arbitrarily – is achievable here: 3 unknowns for exactly 3 equations.

4.3 Pole placement

Criterion for pole selection: all three closed-loop poles are chosen *real and negative*. A complex-conjugate pole pair is what produces oscillation/overshoot in a 2nd-order system (Section 3); by construction, an all-real-pole 3rd-order system has no oscillatory mode, guaranteeing zero overshoot regardless of the specific gains that produce those poles.

A dominant-pole design was used: one slower pole sets the observable response speed, and two much faster poles (whose transients decay quickly and barely influence the visible response) push the system to behave like a well-damped first-order system. Target: an improvement over the open-loop settling time ($t_s \approx 4/(\zeta\omega_n) \approx 637\,\mu s$, Section 3) without demanding unrealistically large gains. Using $t_s \approx 4/|p_{dominant}|$ for a target $t_s \approx 200\,\mu s$:

$$|p_{dominant}| = \frac{4}{200 \times 10^{-6}} = 20{,}000\,\text{rad/s} \tag{19}$$

with the two non-dominant poles placed ∼7.5× further out:

$$p_1 = -20{,}000\,\text{rad/s}, \qquad p_2 = p_3 = -150{,}000\,\text{rad/s} \tag{20}$$

4.4 Gain calculation (coefficient matching)

Expanding the desired closed-loop polynomial from the chosen poles:

$$(s + 20000)(s + 150000)^2 = s^3 + 320000\,s^2 + 28{,}500{,}000{,}000\,s + 4.5 \times 10^{14} \tag{21}$$

Matching term-by-term against Section 4.2's characteristic equation (with $\zeta = 0.2$, $\omega_n \approx 31{,}416\,\text{rad/s}$, $\omega_n^2 \approx 9.8717 \times 10^8$, $2\zeta\omega_n \approx 12566$):

$$s^2: \quad 2\zeta\omega_n + \omega_n^2 K_d = 320000 \;\Rightarrow\; K_d = \frac{320000 - 12566}{9.8717 \times 10^8} \approx 3.114 \times 10^{-4} \tag{22}$$

$$s^1: \quad \omega_n^2(1 + K_p) = 2.85 \times 10^{10} \;\Rightarrow\; K_p = \frac{2.85 \times 10^{10}}{9.8717 \times 10^8} - 1 \approx 27.87 \tag{23}$$

$$s^0: \quad \omega_n^2 K_i = 4.5 \times 10^{14} \;\Rightarrow\; K_i = \frac{4.5 \times 10^{14}}{9.8717 \times 10^8} \approx 455{,}905 \tag{24}$$

$$\boxed{K_p \approx 27.87, \qquad K_i \approx 455{,}905, \qquad K_d \approx 3.114 \times 10^{-4}} \tag{25}$$

Verified. Substituting K_p, K_i, K_d back into the characteristic equation of Section 4.2 reproduces the target polynomial of Section 4.3 to within rounding (e.g. the s^2 coefficient comes out to 319,907 against a target of 320,000, a 0.03% difference from rounding K_d to 4 significant figures).

4.5 Frequency-domain verification

As an independent cross-check (not the design method itself – Sections 4.2–4.4 already place all three poles exactly by construction), the open-loop $L(s) = C(s)G(s)$ and closed-loop $T(s) = \frac{L(s)}{1+L(s)}$ frequency responses were computed numerically (`simulation/bode_plot.py`, using `scipy.signal`, not hand-sketched) from the actual $G(s)$ and $C(s)$ derived above.

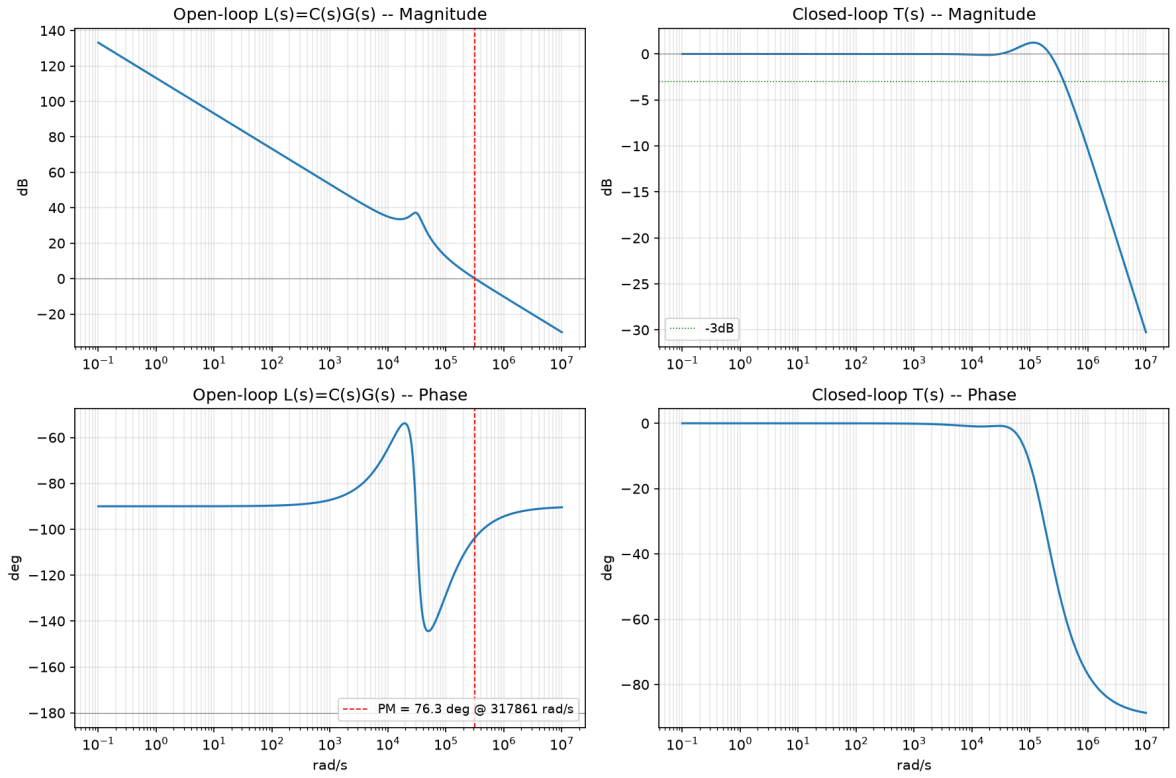


Figure 1: Open-loop $L(s)$ magnitude/phase (left) and closed-loop $T(s)$ magnitude/phase (right), computed from the actual plant and controller transfer functions.

The single gain crossover ($|L(j\omega)| = 0\,\text{dB}$) occurs at $\omega_c \approx 317{,}861\,\text{rad/s}$, with phase margin PM $\approx 76.3°$ – a healthy margin (above the usual 45–60° design target). The open-loop phase never reaches $-180°$ across the swept range (10^{-1} to 10^{7} rad/s), so there is no classical gain-margin crossover to report; this is consistent with the all-real-closed-loop-pole design (Section 4.3), which by construction has no oscillatory mode. The closed-loop magnitude shows a small ($\approx$1 dB) peak near 1–2 $\times\, 10^5$ rad/s before rolling off, and stays flat near 0 dB well below that – consistent with good reference tracking without excessive resonant peaking.

PENDING

Whether the derivative term is implemented as a full analog PID or simplified to PI-only for noise immunity (Section 5) – if K_d is implemented, it must be through the *filtered/band-limited derivative* discussed in Section 5, not an ideal differentiator, since an ideal derivative would amplify any high-frequency noise on the feedback signal.

5 Op-Amp Compensator Circuit

5.1 Topology: five op-amp stages

The parallel PID (Section 4.1) is realized as five op-amp stages: a difference amplifier computing the error, three parallel branches (P, I, filtered-D) acting on that error, and a summing amplifier recombining them.

General basis: the inverting amplifier

Every branch below is a special case of the same ideal-op-amp result (virtual ground at the inverting input + KCL at that node, standard inverting-amplifier analysis):

$$\frac{V_{out}}{V_{in}} = -\frac{Z_f}{Z_{in}} \tag{26}$$

where Z_{in} and Z_f are the input and feedback impedances. Only the choice of components placed in those two positions changes between branches:

1. **Difference amplifier:** computes $e(t) = V_{ref} - V_C$ from the reference and the plant's feedback signal (standard four-resistor differential amplifier). [See Figure 2.]
2. **Proportional branch:** $Z_{in} = R_p$, $Z_f = R_{fp}$:

$$\frac{V_{out,P}}{e} = -\frac{R_{fp}}{R_p} \;\Rightarrow\; K_p = \frac{R_{fp}}{R_p} \tag{27}$$

   [See Figure 4.]
3. **Integral branch:** $Z_{in} = R_i$, $Z_f = \frac{1}{sC_i}$

$$\frac{V_{out,I}}{e} = -\frac{1/(sC_i)}{R_i} = -\frac{1}{sR_iC_i} \;\Rightarrow\; K_i = \frac{1}{R_iC_i} \tag{28}$$

   [See Figure 5.] (division by s in Laplace corresponds to integration in time, hence $V_{out,I} = -\frac{1}{R_iC_i}\int e\,dt$).
4. **Filtered-derivative branch:** $Z_{in} = R_1 + \frac{1}{sC_d}$, $Z_f = R_{fd}$:

$$\frac{V_{out,D}}{e} = -\frac{R_{fd}}{R_1 + \frac{1}{sC_d}} = \frac{-R_{fd}C_d\,s}{1 + sR_1C_d} \tag{29}$$

   giving $K_d = R_{fd}C_d$ at low frequency, capped at $-R_{fd}/R_1$ above the corner frequency $f_{pole} = 1/(2\pi R_1C_d)$ – this avoids amplifying high-frequency noise, unlike an ideal differentiator ($R_1 = 0$, bare capacitor), whose gain grows unboundedly with frequency. [See Figure 6.]
5. **Summing amplifier:** recombines the three branch outputs (each through an equal input resistor R_s, feedback R_{fs}) into $u(t) = K_pe + K_i\int e\,dt + K_d\frac{de}{dt}$, which drives the plant. [See Figure 3.]

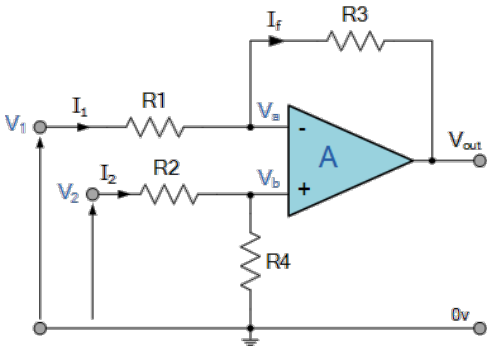


Figure 2: Differential Amplifier

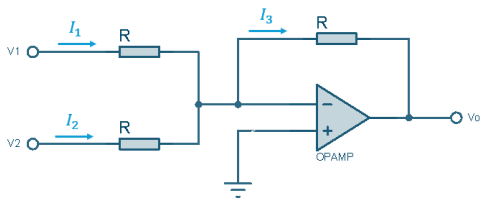


Figure 3: Summing Implementation

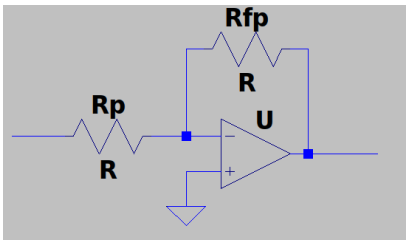


Figure 4: Proportional

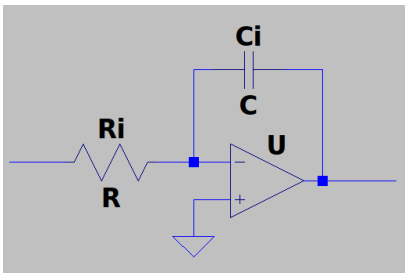


Figure 5: Integral

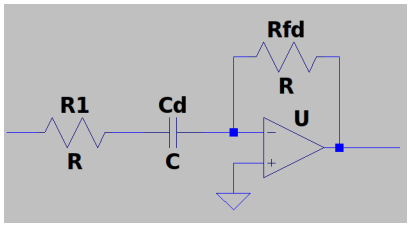


Figure 6: Filtered-Derivative

5.2 Component values

Design choice: $R_p = R_i = R_{fd} = 10\,\text{k}\Omega$ (a common reference value across stages – reduces distinct BOM part count, and $10\,\text{k}\Omega$ is a practical general-purpose value for op-amp input resistors: high enough to not load the driving stage's output, low enough that op-amp input bias current and noise stay negligible). The remaining component in each branch follows directly:

$$R_{fp} = K_p R_p = 27.87 \times 10{,}000 \approx 278.7\,\text{k}\Omega \tag{30}$$

$$C_i = \frac{1}{K_i R_i} = \frac{1}{455{,}905 \times 10{,}000} \approx 219.3\,\text{pF} \tag{31}$$

$$C_d = \frac{K_d}{R_{fd}} = \frac{3.114 \times 10^{-4}}{10{,}000} \approx 31.14\,\text{nF} \tag{32}$$

Filtered-derivative corner frequency and R_1

The corner (pole) frequency of the filtered-derivative branch, in rad/s, is $\omega_{pole} = 1/(R_1 C_d)$ (the pole of $1 + sR_1C_d$). Below ω_{pole} the branch acts as a real differentiator (gain growing with frequency, providing the intended phase lead within the control loop's bandwidth); above it, the gain flattens at a fixed value R_{fd}/R_1 instead of continuing to grow, which is what prevents high-frequency noise from being amplified without bound. Criterion: place ω_{pole} comfortably above the gain crossover frequency ($\approx$ 317,861 rad/s, Section 4.5) so the entire control-relevant frequency range sees real derivative action, with the noise-limiting flattening only taking effect well beyond it. Choosing $\omega_{pole} \approx 2 \times 10^6\,\text{rad/s}$ ($\approx 6.3\times$ the gain crossover):

$$R_1 = \frac{1}{\omega_{pole} C_d} = \frac{1}{(2 \times 10^6)(31.14 \times 10^{-9})} \approx \boxed{16.06\,\Omega} \tag{33}$$

Summing and difference amplifiers

Summing amplifier: each branch (P, I, D) already carries its own gain (K_p, K_i, K_d) from its own stage, so the summer only needs to add the three signals with unity gain, which requires equal input and feedback resistors: $R_s = R_{fs} = 10\,\text{k}\Omega$ (consistent with the rest of the design).

Difference amplifier: the standard 4-resistor differential amplifier gives unity gain ($e = V_{ref} - V_C$ exactly, no scaling) when all four resistors are equal: all four set to $10\,\text{k}\Omega$.

5.3 Full schematic, by stage

Shown one stage at a time (clearer to build from than a single combined diagram). Signal flow: V_{ref} and V_C (feedback) → difference amp → $e(t)$ → P/I/D branches in parallel → summing amp → $u(t)$ → plant (series RLC, Section 3) → V_C, closing the loop.

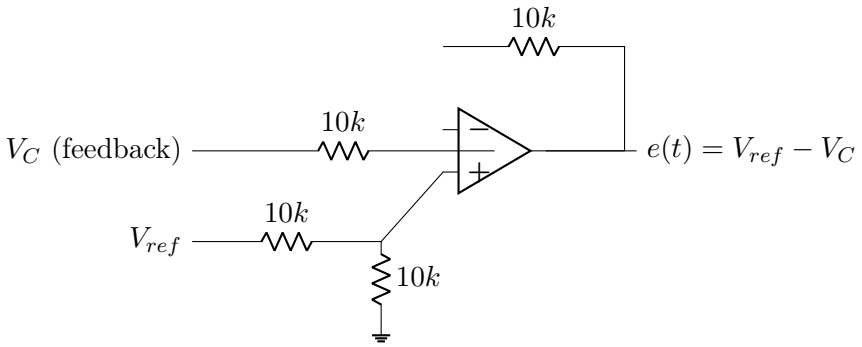


Figure 7: Stage 1 – difference amplifier. All four resistors $10k\Omega$. $V_{ref} \to (+)$, $V_C \to (-)$, giving $e = V_{ref} - V_C$ (not the reverse).

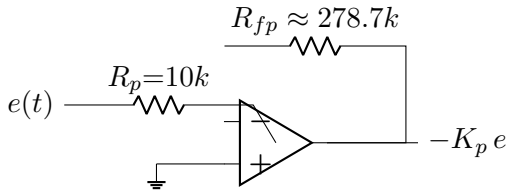


Figure 8: Stage 2 – P branch. Simple inverting amplifier. $K_p = R_{fp}/R_p \approx 27.87$.

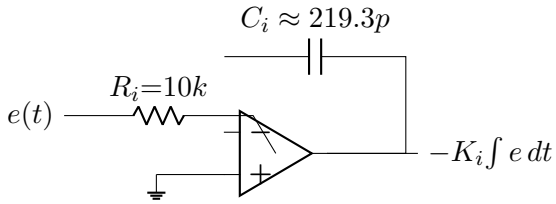


Figure 9: Stage 3 – I branch. Same as P, but a **capacitor** instead of a resistor in the feedback path – that is what makes it an integrator. $K_i = 1/(R_i C_i) \approx 455{,}905$.

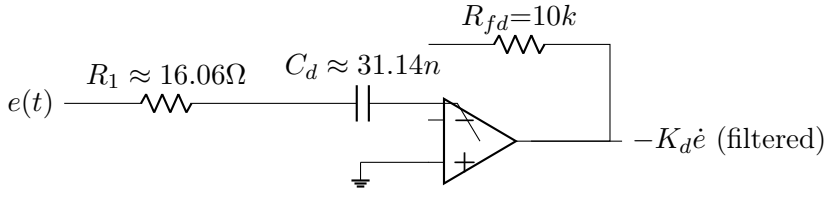


Figure 10: Stage 4 – filtered D branch. The input is R_1 in series with C_d (not a bare capacitor – that would be an ideal, noise-amplifying differentiator). This caps the gain above $f_{pole} = 1/(2\pi R_1 C_d)$.

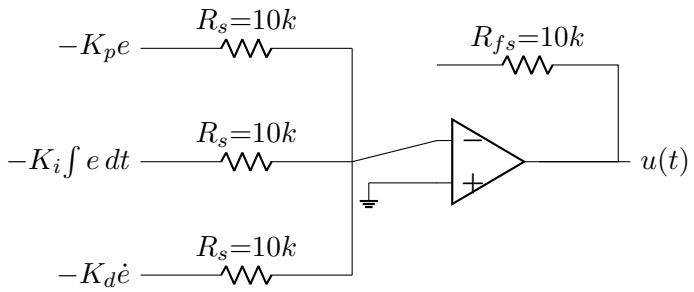


Figure 11: Stage 5 – summing amplifier. The three inputs already carry their own gain; the summer just adds them (unity gain, $R_s = R_{fs}$). Its output $u(t)$ drives the plant (series RLC, Section 3) in place of the step source used for the open-loop test.

Simulated. Built exactly as above in LTspice and run closed-loop, with V_{ref} as a `PULSE(0 1 0 10u 10u 10 10)` source (same 1V step as the open-loop test, 10µs rise time – see Section 6.1 for why an idealized 1ns rise time was rejected). Op-amps modeled with LTspice's `opamp` subcircuit (`.lib opamp.sub`), $A_{ol} = 100k$, GBW raised above the library default of 10MHz to keep each stage's own bandwidth well clear of the loop dynamics (Section 6.2).

Possible failure point: parasitic capacitance on C_i

$C_i \approx 219.3\,\text{pF}$ is small enough that breadboard/PCB stray capacitance (typically 1 pF to 5 pF for a breadboard node) is a non-negligible fraction of it (≈1–2%). If the built integrator's measured K_i deviates noticeably from the calculated 455,905, this parasitic pickup – not necessarily a design or calculation error – should be the first thing checked, before re-deriving the gains. Mitigation if it proves significant: reduce R_i (increasing C_i proportionally to keep the same $K_i = 1/(R_i C_i)$, at the cost of loading the source driving this branch more).

6 SPICE Simulation (LTspice)

6.1 Open-loop

Done – see Section 3.1 (52.68% simulated vs. 52.7% predicted, extracted numerically from the `.raw` file).

6.2 Closed-loop: debugging

Two real issues were hit and resolved before a valid closed-loop result was obtained – documented here since both are genuine, non-obvious LTspice pitfalls, not modeling mistakes in the design itself.

1. Undefined subcircuit. The first run failed with `"This sub-circuit name is not defined"` for all five op-amps. Cause: LTspice's basic `opamp` symbol is not auto-linked to its model – its own description text says *"You must .lib opamp.sub"*. Fix: added `.lib opamp.sub` as a schematic text directive.

2. Numeric overflow ($\sim 4 \times 10^{38}$V). With the missing-library issue fixed, the first successful netlist run produced an output that grew to floating-point overflow – the signature of positive feedback / loop instability, not a slowly-diverging but otherwise sane result. The generated netlist was checked node-by-node against the intended design (Sections 4.4–5) and matched exactly – no wiring error. The actual cause: LTspice's `opamp` model is not truly ideal even at its default settings – it carries a finite gain-bandwidth product, `GBW=10Meg` (10MHz), by default. With $K_p \approx 27.87$,

the P-branch's own bandwidth is only $\mathrm{GBW}/K_p \approx 358\,\mathrm{kHz}$, close enough to the design's own dynamics (D-branch corner $\approx 318\,\mathrm{kHz}$, Section 5) that the cumulative phase lag from five cascaded finite-bandwidth stages ate into the phase margin (Section 4.5) enough to destabilize the loop. Fix: raised `GBW` well above this design's frequencies of interest, confirming the diagnosis (the instability disappeared).

6.3 Closed-loop: result

With both issues resolved, the closed-loop step response was obtained:

Peak V_C	1.115,39 V, at $t = 17.37\,\mu\mathrm{s}$
Final (settled) V_C	0.999,99 V
Overshoot	$(1.11539 - 1)/1 \times 100 = \boxed{11.54\%}$
Settled within ±2%	by $t \approx 31.5\,\mu\mathrm{s}$

This is a large improvement over the 52.68 % open-loop result, but not the 0 % predicted by the all-real-closed-loop-pole design criterion of Section 4.3. Two hypotheses were tested:

Hypothesis A (rejected): step edge speed exciting the D-branch. The reference step's rise time was changed from an idealized 1ns to a more realistic 10 µs (10,000× slower). Overshoot only dropped from 13.26 % to 11.54 %, and the peak timing barely moved (11.3→17.4 µs) – far too small a change for edge speed to be the dominant cause. Rejected by this data.

Hypothesis B (supported): closed-loop zero near the dominant pole. The all-real-pole design criterion (Section 4.3) is sufficient to rule out any oscillatory *pole*, but the closed-loop transfer function also has *zeros*, from the PID's own numerator ($K_d s^2 + K_p s + K_i = 0$) – not accounted for in the original pole-placement criterion. Solving for the actual zeros with the derived gains:

$$K_d s^2 + K_p s + K_i = 0 \;\Rightarrow\; z_{1,2} \approx -21{,}524 \text{ and } -67{,}979 \text{ rad/s} \tag{34}$$

$z_1 \approx -21{,}524$ rad/s sits almost on top of the dominant pole ($p_1 = -20{,}000$ rad/s, Section 4.3) – a near pole-zero cancellation. When a zero nearly cancels the dominant (slowest) pole, the visible response is no longer dominated by that $\sim 200\,\mu\mathrm{s}$ time constant – it is dominated by the remaining, much faster dynamics (the non-dominant poles at $-150{,}000$ rad/s, $\tau \approx 6.7\,\mu\mathrm{s}$, and the second zero at $z_2 \approx -67{,}979$ rad/s, $\tau \approx 14.7\,\mu\mathrm{s}$), which matches the observed 11–31 µs transient far better than the design's nominal 200 µs target. A real pole-zero pair (both on the negative real axis, not complex) can still produce step-response overshoot when imperfectly cancelled – a real, documented limitation of the pole-placement method used, not a simulation artifact.

Decision: the 11.54 % result is accepted as final for this project phase, given the trade-off explored in Section 6.4.

6.4 Pole placement revisited: why not just re-tune?

A systematic search (varying the dominant pole magnitude and its separation ratio from the non-dominant poles, and separately trying a complex dominant pole pair with $\zeta = 0.7$) found that a zero landing near the dominant pole is a persistent, structural feature of this plant-plus-PID combination, not a poor choice of one specific pole triple: every fast/practical candidate tested still put a zero within 4–25 % of the dominant pole. Materially improving that separation required slowing the dominant pole down (e.g. to $\sim$5000 rad/s, $\tau \approx 800\,\mu\mathrm{s}$) – slower than the *open-loop* response itself, defeating the purpose of adding control. This trade-off (speed vs. overshoot, given this specific plant) is the actual reason 11.54 % was accepted rather than chased further.

7 Conclusions

1. The open-loop plant matched its analytical prediction almost exactly (52.68 % simulated vs. 52.7 % calculated overshoot), validating both the transfer-function derivation (Section 3) and its SPICE implementation before any control was added.

2. A parallel PID, designed by direct algebraic pole placement (3 free gains for a 3rd-order closed-loop characteristic equation, Section 4), reduced the overshoot from 52.68 % (open-loop) to 11.54 % (closed-loop) – a large, real improvement, though not the 0 % the all-real-pole criterion predicted.

3. **Key limitation found:** an all-real-pole closed-loop design guarantees no oscillatory *pole*, but does not by itself guarantee zero step-response overshoot, because the PID contributes its own *zeros* to the closed-loop transfer function. For this plant, a zero consistently lands near the dominant pole across a wide range of pole choices (Section 6.5) – a structural property of this specific plant-plus-PID combination, confirmed by systematically searching the design space, not a one-off tuning mistake.

4. Two real LTspice pitfalls were hit and resolved during closed-loop debugging: the generic `opamp` symbol requires a manual `.lib opamp.sub` directive, and that same symbol's default gain-bandwidth product (10MHz) was not fast enough for this specific design (five cascaded stages, $K_p \approx 27.87$) without destabilizing the loop – both documented in Section 6.2 so they are not rediscovered from scratch on the next project (the PID-controlled buck).

5. The op-amp compensator was fully specified with real, standard component values ($R_p = R_i = R_{fd} = R_s = R_{fs} = 10\,\text{k}\Omega$, $R_{fp} \approx 278.7\,\text{k}\Omega$, $C_i \approx 219.3\,\text{pF}$, $R_1 \approx 16.06\,\Omega$, $C_d \approx 31.14\,\text{nF}$, Section 5), and the filtered-derivative design (a lead network, not an ideal differentiator) was carried through consistently to avoid amplifying switching/measurement noise.

6. 11.54 % overshoot, well-characterized and understood, is accepted as the final result for this project phase rather than chased further, given the documented speed-vs-overshoot trade-off (Section 6.5).